

OPTIMIZING WALMART'S RETAIL PERFORMANCE

Data-Driven Decision Support Analysis

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Technologies: SQLite | SQL | Power BI

Executive Summary

This analysis examines 6,435 weekly sales records across 45 Walmart stores from February 2010 to October 2012. Six stores contribute over 25% of total company revenue, while 18 stores operate below 80% of the company-wide average. Holiday periods generate a 7.84% average sales uplift, yet five stores experience negative holiday growth—a controllable operational failure. Macroeconomic factors show weak explanatory power, confirming that store-level execution drives performance variance. The analysis concludes with a three-tier investment framework and a 30-60-90 day action plan for the Regional Operations Director.

Business Scenario & Problem

A newly appointed Regional Operations Director must allocate finite management bandwidth and capital across 45 stores. Without systematic benchmarking, resources risk misallocation to minor issues while critical underperformers deteriorate. The organization lacks empirical evidence on whether performance gaps stem from controllable operational factors or uncontrollable external conditions. Seasonal demand patterns remain poorly understood at the store level, preventing proactive capacity planning.

Objectives

1. **Performance Prioritization** — Identify and classify stores by intervention urgency.
2. **Performance Diagnosis** — Determine drivers of sales variance across the network.
3. **Investment Optimization** — Direct capital, staffing, and inventory toward maximum-ROI opportunities.

Dataset Overview

The dataset contains 6,435 weekly observations from 45 Walmart stores (Feb 2010 – Oct 2012). Variables include Store ID, Date, Weekly_Sales, Holiday_Flag, Temperature, Fuel_Price, CPI, and Unemployment. Data quality validation confirmed zero missing values, zero duplicates, and zero invalid records.

Executive Question 1: Which Stores Require Immediate Attention?

Analysis 1.1: Store Performance Benchmarking

The chart below ranks all 45 stores by cumulative revenue. Store 20 leads at ~\$301M, while Store 33 trails at ~\$37M—a performance spread of 8.1x. This disparity indicates structural differences in market size or operational efficiency, not random variation.

[Chart: Sum of Weekly_Sales by Store — horizontal bar chart ranking all 45 stores]

Key Finding: Six stores contribute over 25% of total revenue. The bottom 10 stores each generate less than \$100M in cumulative sales.

Analysis 1.2: Persistent Underperformance

Twenty-seven of 45 stores (60%) consistently operate below the company-wide average weekly sales of \$1,046,965. Store 33 operates at just 25% of benchmark. This is structural underperformance requiring immediate intervention, not passive monitoring.

Analysis 1.3: Operational Priority Classification

Given limited operational resources, stores were classified into four priority tiers based on performance ratio relative to the company average.

Priority Level	Store Count	Criteria	Action
Critical	9	< 50% of average	Immediate diagnostic review
High	8	50% – 75% of average	Targeted improvement plan
Medium	10	75% – 100% of average	Monitor with selective support
Low	18	>= 100% of average	Maintain and benchmark

Interpretation: Nearly 40% of the network requires active intervention. The 18 Low-Priority stores should serve as internal benchmarks for operational best practices.

Store Performance Rankings

The following tables provide a detailed breakdown of the highest and lowest performing stores by total cumulative revenue over the analysis period.

Top 10 Best Performing Stores

Rank	Store	Total Sales
1	20	\$301.40M
2	4	\$299.54M
3	14	\$288.99M
4	13	\$286.52M
5	2	\$275.38M
6	10	\$271.62M
7	27	\$253.86M
8	6	\$223.76M
9	1	\$222.40M
10	39	\$207.45M

Bottom 10 Performing Stores

Rank	Store	Total Sales
1	33	\$37.16M
2	44	\$43.29M
3	5	\$45.48M
4	36	\$53.41M
5	38	\$55.16M
6	3	\$57.59M
7	30	\$62.72M
8	37	\$74.20M
9	16	\$74.25M

10	29	\$77.14M
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Insight: The revenue gap between the #1 store (Store 20 at \$301.40M) and the lowest-performing store (Store 33 at \$37.16M) represents an 8.1x performance spread, confirming the need for tiered resource allocation and targeted operational intervention.

Executive Question 2: Why Are Some Stores Outperforming Others?

Analysis 2.1: Holiday Sales Impact

The chart below compares total revenue between non-holiday and holiday weeks. While non-holiday weeks dominate in absolute terms, holiday weeks generate a 7.84% per-week sales uplift.

[Chart: Sum of Weekly_Sales by Holiday_Flag]

Week Type	Avg Weekly Sales	Total Weeks
Holiday Week	\$1,122,887.89	450
Non-Holiday Week	\$1,041,256.38	5,985

Key Finding: Holidays drive meaningful but uneven demand. Company-wide averages mask significant store-level variation.

Analysis 2.2: Store-Level Holiday Performance

Holiday demand is not evenly distributed. Store 7 achieves a 19.44% holiday lift, while five stores actually lose revenue during holidays—a counterintuitive finding suggesting inventory stockouts, understaffing, or poor promotional execution during peak periods.

Category	Store	Holiday Lift	Implication
Top Performer	Store 7	+19.44%	Replicate strategies
Underperformer	Store 44	-2.38%	Operational audit needed

Recommendation: Audit holiday operations at the five negative-lift stores immediately. These are controllable failures, not market conditions.

Analysis 2.3: External Factors and Sales Performance

Macroeconomic factors explain minimal variance in sales performance. Temperature and fuel price show virtually no difference between high and low sales periods. CPI and unemployment differences are statistically present but practically insignificant.

Factor	High-Sales Weeks	Low-Sales Weeks
CPI	170.71	172.28
Unemployment	7.86%	8.11%

Key Insight: Store-level operational execution drives performance far more than local economic conditions. Management cannot blame the economy for underperformance.

Executive Question 3: Where Should Walmart Invest Resources?

Analysis 3.1: Revenue Contribution Analysis

Stores were classified into three investment tiers based on revenue contribution.

Investment Tier	Count	Revenue Share	Strategy
Strategic Investment	6	>= 4.0% each	Protect and expand
Growth Opportunity	11	2.5% – 4.0% each	Accelerate growth
Maintenance	28	< 2.5% each	Efficiency only

Interpretation: The top 6 stores generate disproportionate returns and warrant protected investment. The 11 Growth Opportunity stores offer the highest marginal ROI.

Analysis 3.2: Monthly Sales Trend Analysis

The charts below reveal highly predictable seasonal patterns. December peaks at ~\$1.28M average weekly sales, while January troughs at ~\$909K—a 40.8% seasonal swing. The yearly trend shows growth from 2010 to 2011, followed by a sharp 2012 decline requiring investigation.

[Charts: Sum of Weekly_Sales by Month & Sum of Weekly_Sales by Year]

Quarterly view confirms Q2 and Q3 as the strongest periods, with Q1 consistently weakest.

[Chart: Sum of Weekly_Sales by Quarter]

Recommendation: Increase December inventory by 40% and staffing by 25% in Strategic and Growth stores. Reduce January inventory by 20% network-wide.

Analysis 3.3: Executive KPI Dashboard

The KPI cards below provide an at-a-glance summary of network performance. These metrics should form the foundation of a weekly executive review.

KPI	Value	Target
Total Revenue	\$6.74B	Grow YoY
Avg Weekly Sales	\$1.05M	Maintain or grow
Performance Spread	18.2x	< 10x by 2014

Strategic Recommendations

Immediate Actions (0–30 Days)

1. Deploy Rapid Response Teams to 9 Critical Priority stores (33, 44, 5, 36, 38, 3, 30, 37, 16) for diagnostic audits.
2. Audit holiday operations at 5 negative-lift stores (44, 36, 38, 37, 30) for stockouts, staffing gaps, and promotional failures.
3. Implement weekly KPI alerts flagging any store >10% below its 4-week rolling average.

Short-Term Investments (1–3 Months)

4. **Seasonal Capacity Scaling:** Increase December inventory by 40% and staffing by 25% in Strategic and Growth stores. Reduce January inventory by 20% network-wide.
5. **Benchmark Replication:** Document protocols from Store 20 (highest revenue) and Store 7 (best holiday execution) for network rollout.

Strategic Investments (3–12 Months)

6. **Portfolio Rationalization:** Evaluate lease renewals for bottom 10 stores contributing <1% revenue each.
7. **Predictive Analytics:** Build forecasting models using holiday flag, seasonal month, and store tier to optimize labor scheduling.
8. **Power BI Dashboard:** Deploy four-page executive dashboard refreshed weekly.

Conclusion

Walmart's 45-store network exhibits structural performance variation driven primarily by operational execution—not macroeconomic conditions. The 9 Critical Priority stores and 5 holiday underperformers represent controllable revenue leaks requiring immediate intervention. The 6 Strategic Investment stores and 11 Growth Opportunity stores should receive protected investment. Predictable seasonal patterns (40.8% December peak) enable proactive capacity planning. By replacing uniform management with tiered, data-driven resource allocation, the Regional Operations Director can transform network performance from a liability into a competitive advantage.